Abderrahmene Naceri

Project Portfolio | Robotics, Embedded and FPGA Systems

naceri@uwindsor.ca (226) 246-0994 linkedin.com/in/abderrahmene-naceri github.com/ANaceri26 anaceri26.github.io Windsor, ON

Electrical Engineering student at the University of Windsor. I lead the **robotic arm on the UWindsor Rover Team**, write embedded software for **Lancer Motorsports Formula SAE**, and build FPGA and embedded projects on the side. Every title below is a link to its page.

SHORT ON TIME?

- **Rover arm controls →** ROS 2, MoveIt 2, Gazebo, ODrive bring-up
- **I2C controller core →** SystemVerilog RTL, C driver, on hardware
- **Edge AI classifier →** int8 MobileNetV2 on an ESP32-S3

CLUBS AND TEAMS			PAGE
UWindsor Rover Team: Robotic Arm	Arm Team Lead: control software, simulation, motor bring-up	Robotics · C++ · ROS 2	2
Lancer Motorsports: Formula SAE	CAN/FDCAN, throttle and brake sensing, dashboard I/O	Firmware · STM32 · CAN	2
FPGA AND DIGITAL DESIGN			
VGA Controller and Frame Buffer	640×480 video pipeline from block RAM on a MAX 10	Verilog · Quartus	3
I2C Controller, Driver and Arduino Link	Memory-mapped RTL, bare-metal C driver, 3.3 V–5 V link	SystemVerilog · C · Python	3
DMA Controller and Arbiter	32-bit copy engine sharing a memory port with a CPU	SystemVerilog · Verilator	3
Ripple-Carry and Lookahead Adders	Gate-level adders, binary and decimal display on a DE10-Lite	Verilog · Questa	3
EMBEDDED AND EDGE AI			
Edge AI Cat/Dog Classifier	Quantized MobileNetV2 running on a camera board	TensorFlow · TFLite Micro	4
Bluetooth and Wi-Fi Signal Finder	Handheld RSSI tracker with a live heatmap app	C++ · NimBLE · Python	4
Autonomous Fertilizer Robot	Line following, obstacle avoidance, dispenser; finalist	Arduino · C++ · CAD	4
Solar-Powered Vehicle	Two-motor drive charged from two solar panels	Power · H-bridge	4
COMPETITIONS AND SYSTEMS			
Interstellar Highway	Orbital trade-route planner, Ontario Engineering Competition	Python · orbital mechanics	5
ICTC CyberTitan	3rd place in Ontario, defending systems under attack	Windows Server · Wireshark	5
Home Server and Security Lab	Media server, cloud, DNS ad blocking, VPN, security lab	Linux · networking	5
More: simulation, CAD, software	Simulink, AutoCAD GD&T, Python login system, VBA grading	MATLAB · AutoCAD · VBA	5

Running on hardware · Verified in simulation · Competition · Team role

SKILLS

Digital design
Verilog, SystemVerilog, Intel Quartus, ModelSim/Questa, Icarus Verilog, Verilator, Yosys, MAX 10 / DE10-Lite, FSMs, self-checking testbenches, SDC timing constraints

Embedded and robotics
C, C++, STM32 Nucleo-H723ZG, ESP32-S3, Arduino, Raspberry Pi/Pico, Jetson Orin NX; CAN/FDCAN, I2C, SPI, UART, ADC, DMA; ROS 2, MoveIt 2, Gazebo, MuJoCo, ODrive S1

Software, ML and systems
Python, MATLAB/Simulink, Bash, VBA; TensorFlow, TFLite Micro, int8 quantization; Ubuntu Server, Linux CLI, DNS, VPN, Kali Linux, Nmap, Wireshark

Lab, mechanical and languages
Oscilloscopes, multimeters, soldering, AutoCAD, GD&T (ASME Y14.5), 3D printing. Spoken: English, French, Arabic, Spanish

Videos of these running are on my website. **anaceri26.github.io →**

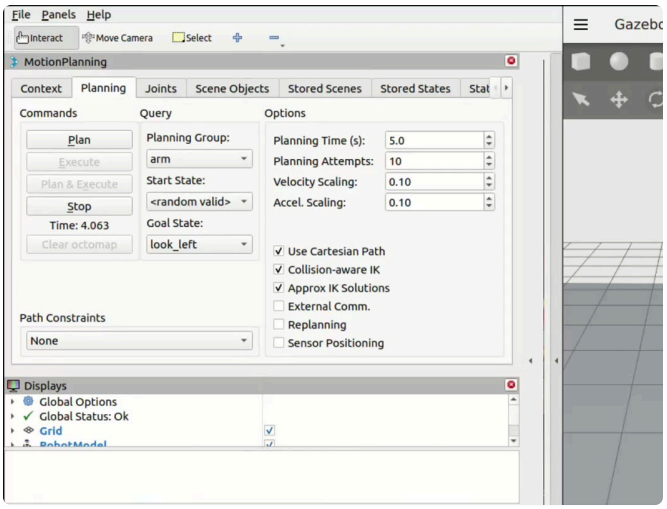
Clubs and teams

UWindsor Rover Team: Robotic Arm

Jan 2026 – present

Arm Team Lead, Robotics Controls and Software

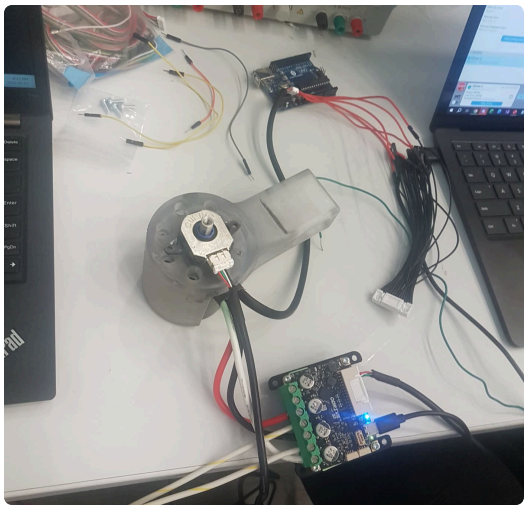
C++, Python, ROS 2 Humble, MoveIt 2, Gazebo, ros2_control, URDF, MuJoCo, Simulink, Jetson Orin NX, Arduino, ODrive S1



MoveIt 2 plans in RViz and executes in Gazebo through ros2_control.



The rover and arm.



ODrive S1 bring-up with an absolute encoder.

Jetson Orin NX	ROS 2 Humble	5 joints + gripper	ODrive S1
arm computer + Arduino actuator layer	MoveIt 2, ros2_control, URDF, RViz	arm model built from team CAD	REV NEO v2.1 BLDC

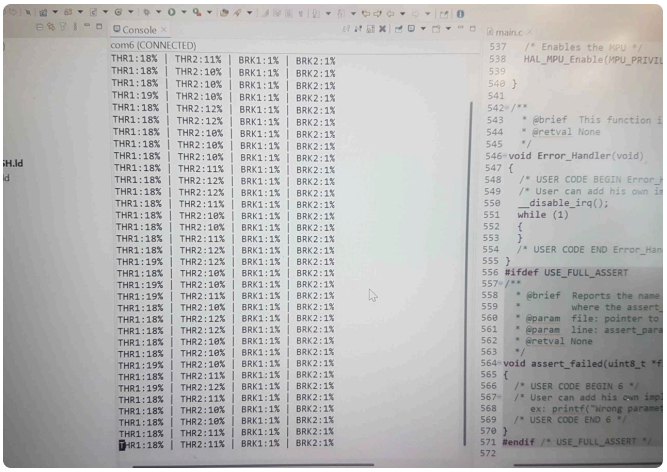
- C++ command interface mapping keyboard and Thrustmaster T.16000M joystick input to multi-axis arm motion, with joint-limit and collision-safety logic that blocks unsafe commands.
- Jetson Orin NX runs high-level commands; an Arduino layer drives the low-level actuators.
- Gazebo Fortress simulation built from the team CAD (collision shapes, rover-body envelope); MuJoCo workspace and manipulability analysis; Simulink drivetrain models.
- ODrive S1 bring-up with a REV NEO v2.1 BLDC, evaluating Hall and AMT212B-OD absolute-encoder feedback and fixing velocity-limit and feedback faults on the test board.

Lancer Motorsports: Formula SAE

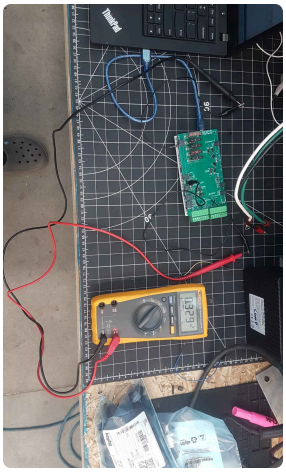
Jan 2026 – present

Embedded Software and Controls

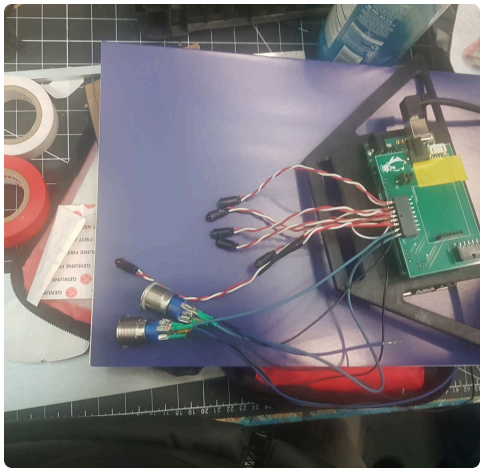
C, Python, STM32 Nucleo-H723ZG, STM32CubeIDE, Arduino Uno/Mega, MCP2515, Raspberry Pi/Pico, Simulink, oscilloscope



Throttle and brake (THR1/THR2, BRK1/BRK2) streamed over UART from the Nucleo-H723ZG.



Bench-testing a team PCB.



Dashboard I/O bench test.

CAN + FDCAN	12-bit	3 MCU families	Simulink
inverter, BMS, dashboard, fault traffic	multi-ADC throttle/brake, circular DMA	Uno, Mega, H723ZG via MCP2515	VCU models before hardware

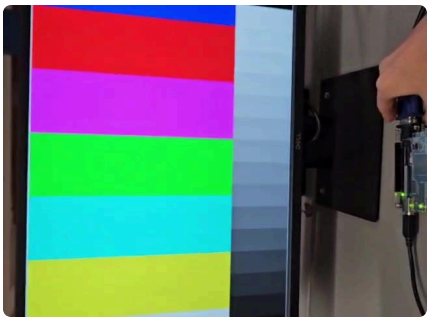
- CAN read/write over SPI with MCP2515 interfaces on Arduino Uno, Mega and Nucleo-H723ZG, including frame, wiring and baud-rate debugging; classic CAN and FDCAN with periodic transmit intervals.
- Multiple ADC peripherals for throttle and brake with channel ranking and circular DMA; resolved sampling and DMA conflicts via sample times, clocking and stream assignments.
- Reverse-engineered dashboard I/O (I2C expanders, 7-segment drivers); checked PCB traces with an oscilloscope; Raspberry Pi script to offload test data to USB.

FPGA and digital design

VGA Timing Controller and Frame Buffer

Running on hardware

Verilog, Intel Quartus, ModelSim, Icarus Verilog, DE10-Lite (MAX 10)



Test card on a monitor, driven from block RAM.

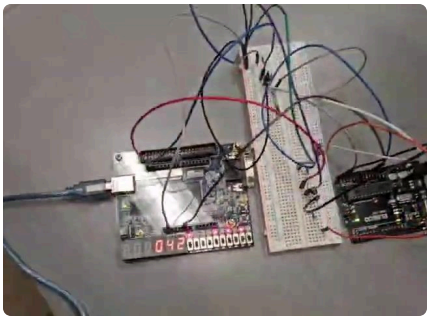
640×480 @ 60 Hz	921,600 bits	307,200 px
25 MHz pixel enable	dual-port BRAM buffer	checked per frame

- Counter-based timing generator with active-low sync; one 50 MHz clock domain, two-flop switch synchronizers, SDC constraints.
- 320×240 RGB444 frame buffer in M9K RAM, upscaled 2×2; painter writes in vertical blanking so frames never tear.
- Self-checking testbench rebuilds position from sync pulses and checks every visible pixel across six frames (~5.8 M cycles).

I2C Controller Core, C Driver and FPGA-to-Arduino Link

Running on hardware

SystemVerilog, C11, C++ (Arduino), Icarus Verilog, Verilator, Yosys, Python, DE10-Lite, Arduino Uno, BS170



DE10-Lite reading 042 from an Uno through the level shifter.

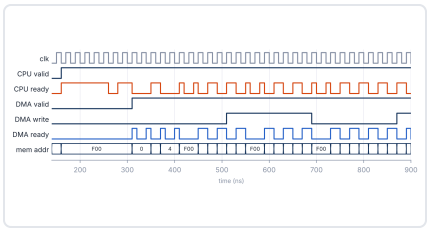
18-state FSM	1,000 × 2 seeds	0xE5
START, rep. START, ACK/NACK, stretching	random transactions, 0 failures	ADXL345 ID read on the board

- Memory-mapped controller: open-drain outputs, input synchronizers, watchdog, W1C status and IRQ; C11 HAL runs on real MMIO or against Verilated RTL (1,600-byte co-sim).
- DE10-Lite demo reads the onboard ADXL345 and lights a pass LED; a second build talks to an Uno at 10 kHz over a BS170 3.3 V–5 V shifter.
- Found Quartus silently ignoring all 72 pin assignments; fixed the QSF and wrote a Python pinout checker.

DMA Controller with Round-Robin Bus Arbiter

Verified in simulation

SystemVerilog, C11, Verilator, Icarus Verilog, Yosys



CPU stand-in and DMA alternating beats on the shared port.

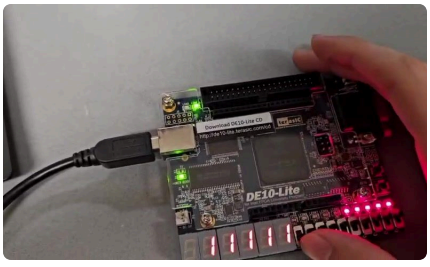
1–16 words	260,000+	500
burst length	contended bus beats verified	C driver co-sim copies

- 32-bit copy engine with CSRs, IRQ, error and timeout reporting; rejects misaligned, overlapping and overflowing copies in hardware.
- Fair beat-level round robin that holds a grant across stalls, keeping data stable under backpressure.

4-Bit Ripple-Carry and Carry-Lookahead Adders

Running on hardware

Verilog, Intel Quartus, Questa, DE10-Lite (MAX 10)



1111 + 1111 + carry-in = 31.

4-bit	0–31	C1–C4
A + B + carry-in	binary or decimal readout	lookahead from A, B, C0 only

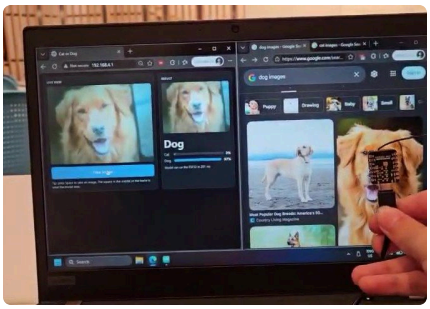
- Ripple-carry from chained half-adder stages vs carry-lookahead with generate/propagate terms, removing the ripple chain.
- Switch inputs, LED sum and carry, and a 7-segment toggle between binary and decimal; pins mapped in Pin Planner.

Embedded and edge AI

ESP32-S3 Edge AI Cat/Dog Classifier

Running on hardware

Python, TensorFlow, TensorFlow Lite Micro, C++, Seeed XIAO ESP32-S3 Sense



The board reporting "Dog" from its camera.

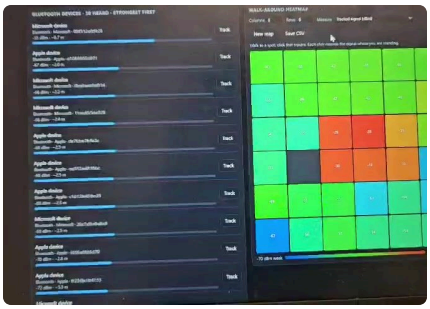
23,262	96×96 int8	1 MB	1.5 s
training images	MobileNetV2 (0.35)	PSRAM tensor arena	per prediction

- Transfer learning: new head at 1e-3, then top 40 layers at 1e-5 with BatchNorm frozen; full int8 quantization with a 300-image calibration set, accuracy re-checked in int8.
- Model compiled into flash as a C array; TFLite Micro with ESP-NN; frames center-cropped exactly as in training.
- Results served on the board's own Wi-Fi page; under 70% confidence it reports "not sure".

ESP32 Bluetooth and Wi-Fi Signal Finder

Running on hardware

C++, Python, NimBLE, UDP, JavaScript, Seeed XIAO ESP32-S3



Walk-around heatmap from live Bluetooth scans.

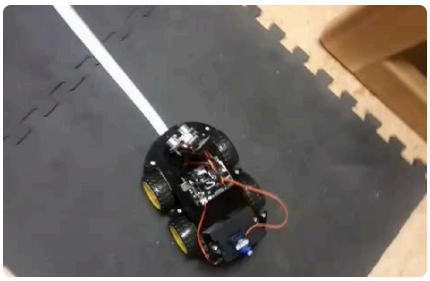
80	300 ms	α = 0.3	400 ms
devices tracked	UDP report interval	RSSI smoothing	single-channel rescan

- Scans BLE advertisements and Wi-Fi beacons, expires stale devices after 20 s, streams the 30 strongest over UDP.
- Buzzer rate and pitch rise with signal strength, even out of laptop range; BOOT button switches modes or locks on.
- Standard-library Python app: live list, "get closer" tracking and a click-to-record RSSI heatmap with CSV export.

Autonomous Fertilizer Robot

Competition finalist

C++, Arduino Uno (ATmega328P), ultrasonic and line sensors, AutoCAD, LEGO



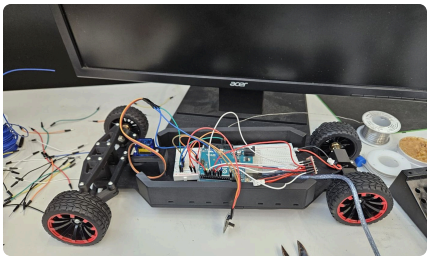
Test run before the competition.

< 40 s	$32.45	Finals
full course time	build cost	competition result

- Line tracking plus ultrasonic obstacle avoidance; code tightened to fit the ATmega328P's memory for the finals.
- Modular LEGO dispenser; AutoCAD used to plan weight balance and fertilizer delivery.

Solar-Powered Vehicle

DC motors, dual H-bridge, solar charge controller, batteries, Arduino Uno



Chassis with servo steering, driver and Uno.

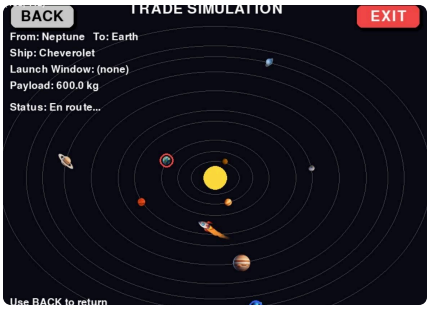
- Both motors bench-tested through a dual H-bridge on independent channels before permanent wiring.
- Two solar panels through a charge controller to the battery; output verified under flashlight and sunlight.
- Soldered power and control wiring, driver-enable logic and dual-battery integration; IR remote control decoded on the Uno.

Competitions and systems

Interstellar Highway

Ontario Engineering Competition

Python, JSON, orbital mechanics (Lambert's problem, Tsiolkovsky rocket equation)



Trade simulation: Neptune to Earth, 600 kg payload.

10	7	83%
celestial bodies	spacecraft classes	trajectory accuracy vs theory

- Lambert solvers compute transfer orbits; the rocket equation gives delta-v and fuel per ship class, optimizing launch windows and payload.
- Ranks trade routes and returns flight data as JSON, with an animated simulator, CLI and GUI.

ICTC CyberTitan

3rd place, Ontario

Windows Server, Remote Desktop, Wireshark, WinPcap, Local Security Policy, firewall rules



Hardening systems during the competition.

3rd	15+
province-wide	vulnerabilities found and fixed

- Kept a simulated lunar base running under cyberattack by auditing system logs and firewall rules under time pressure.
- Reviewed Windows Local Security Policy and firewall configuration to cut the attack surface.

Home Server and Security Lab

Running at home

Linux, Raspberry Pi, Ubuntu Server, Bash, DNS, VPN, Plex, Kali Linux, Nmap, Wireshark, Metasploit



The Pi with its storage drive.

- Always-on Raspberry Pi: Plex streaming, personal cloud storage replacing a paid plan, and a self-hosted VPN back home.
- The Pi is the network's DNS server on a static IP, blocking ads and trackers for every device.
- Isolated Kali/Ubuntu VM lab: Nmap, Wireshark and Metasploit used to find and fix misconfigurations in legacy routing hardware.

More projects

Dynamic Systems Simulation

MATLAB, Simulink

y'' + 6y' + 4y = 2 built from integrators and feedback; simulated steady state matched the hand-solved 0.5.

Mechanical Assembly Design

AutoCAD, GD&T (ASME Y14.5)

Full drawing set for a 1020 steel bracket assembly: orthographic, section and isometric views with a BOM.

Secure Login System

Python, Tkinter, hashlib

SHA-256 password hashing, CSV user storage, 30-second lockout after 3 failed attempts, themed GUI.

Automatic Grades Calculator

Excel, VBA

Macros compute weighted grades and build a worksheet per class from imported student data.